

Labsheet-2

1.Student Registration Form: Create a Student Registration Form using HTML. Include text fields, radio buttons, checkboxes, dropdown list, textarea, submit, and reset buttons. Apply CSS to style the form.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Student Registration Form</title>
```

```
  <style>
```

```
    body{
```

```
      font-family: Arial, sans-serif;
```

```
      background-color: #f2f2f2;
```

```
    }
```

```
    .container {
```

```
      width: 500px;
```

```
      margin: 30px auto;
```

```
      background-color: white;
```

```
      padding: 25px;
```

```
      border-radius: 10px;
```

```
      box-shadow: 0 0 10px gray;
```

```
    }
```

```
    h1 {
```

```
      text-align: center;
```

```
    color: #333;  
}
```

```
label {  
    font-weight: bold;  
}
```

```
input[type="text"],  
input[type="email"],  
input[type="password"],  
input[type="date"],  
select,  
textarea {  
    width: 100%;  
    padding: 10px;  
    margin-top: 5px;  
    margin-bottom: 15px;  
    border: 1px solid #ccc;  
    border-radius: 5px;  
    box-sizing: border-box;  
}
```

```
.gender,  
.hobbies {  
    margin-bottom: 15px;  
}
```

```
input[type="submit"],  
input[type="reset"] {  
    padding: 10px 25px;  
    border: none;  
    border-radius: 5px;  
    cursor: pointer;  
    margin-right: 10px;  
}
```

```
input[type="submit"] {  
    background-color: green;  
    color: white;  
}
```

```
input[type="reset"] {  
    background-color: red;  
    color: white;  
}
```

```
input[type="submit"]:hover,  
input[type="reset"]:hover {  
    opacity: 0.8;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>Student Registration Form</h1>
```

```
<form>
```

```
<!-- Text Field -->
```

```
<label>Name:</label>
```

```
<input type="text" name="name" placeholder="Enter your name">
```

```
<!-- Email Field -->
```

```
<label>Email:</label>
```

```
<input type="email" name="email" placeholder="Enter your email">
```

```
<!-- Password Field -->
```

```
<label>Password:</label>
```

```
<input type="password" name="password" placeholder="Enter password">
```

```
<!-- Radio Buttons -->
```

```
<label>Gender:</label>
```

```
<div class="gender">
```

```
<input type="radio" name="gender" value="male"> Male
```

```
<input type="radio" name="gender" value="female"> Female
```

```
<input type="radio" name="gender" value="other"> Other
</div>
```

```
<!-- Date Field -->
<label>Date of Birth:</label>
<input type="date" name="dob">
```

```
<!-- Dropdown List -->
<label>Course:</label>
<select name="course">
  <option value="">-- Select Course --</option>
  <option value="bca">BCA</option>
  <option value="bba">BBA</option>
  <option value="bcom">B.Com</option>
  <option value="btech">B.Tech</option>
</select>
```

```
<!-- Checkboxes -->
<label>Hobbies:</label>
<div class="hobbies">
  <input type="checkbox" name="hobby" value="reading"> Reading
  <input type="checkbox" name="hobby" value="coding"> Coding
  <input type="checkbox" name="hobby" value="sports"> Sports
  <input type="checkbox" name="hobby" value="music"> Music
</div>
```

```
<!-- Textarea -->  
  
<label>Address:</label>  
  
<textarea name="address" rows="5"  
    placeholder="Enter your address"></textarea>
```

```
<!-- Submit and Reset Buttons -->  
  
<input type="submit" value="Submit">  
  
<input type="reset" value="Reset">
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

2.College Website using Frames: Create a college website using HTML Frames with Header, Navigation Menu, and Main Content sections. Add navigation links that display different pages in the main frame.

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
  
    <title>College Website</title>  
  
</head>
```

```
<frameset rows="20%,80%">
```

```
<!-- Header and Main Section -->
```

```
<frameset cols="100%">

  <frame src="header.html" name="header">

</frameset>
```

```
<frameset cols="25%,75%">

  <!-- Navigation Menu -->

  <frame src="menu.html" name="menu">


  <!-- Main Content -->

  <frame src="home.html" name="main">

</frameset>
```

```
</frameset>
```

```
</html>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Header</title>
```

```
</head>
```

```
<body style="background-color: lightblue; text-align: center;">
```

```
  <h1>COER University</h1>
```

```
  <h3>Welcome to Our College Website</h3>
```

```
</body>
```

```
</html>

<!DOCTYPE html>

<html>

<head>

  <title>Navigation Menu</title>

</head>

<body>


  <h2>Navigation Menu</h2>


  <ul>

    <li>

      <a href="home.html" target="main">Home</a>

    </li>


    <li>

      <a href="about.html" target="main">About Us</a>

    </li>


    <li>

      <a href="courses.html" target="main">Courses</a>

    </li>


    <li>

      <a href="contact.html" target="main">Contact Us</a>

    </li>
```


</body>

</html>

<!DOCTYPE html>

<html>

<head>

<title>Home</title>

</head>

<body>

<h1>Home</h1>

<p>

Welcome to COER University. Our college provides quality education
and a supportive learning environment for students.

</p>

</body>

</html>

<!DOCTYPE html>

<html>

<head>

<title>About Us</title>

</head>

<body>

<h1>About Us</h1>

<p>

COER University is committed to providing quality education,
practical knowledge, and opportunities for students to develop
their skills.

</p>

</body>

</html>

<!DOCTYPE html>

<html>

<head>

<title>Courses</title>

</head>

<body>

<h1>Courses Offered</h1>

Bachelor of Computer Applications (BCA)

Bachelor of Business Administration (BBA)

Bachelor of Commerce (B.Com)

Bachelor of Technology (B.Tech)

</body>

</html>

<!DOCTYPE html>

<html>

<head>

 <title>Contact Us</title>

</head>

<body>

 <h1>Contact Us</h1>

 <p>

 College: COER University

 </p>

 <p>

 Email: info@college.com

 </p>

 <p>

 Phone: 9876543210

 </p>

</body>

</html>

3.Login and Admission Form: Design a Login Form and College Admission Form using HTML. Use appropriate form elements and apply CSS for formatting, borders, spacing, fonts, and buttons.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Login and College Admission Form</title>
```

```
  <style>
```

```
    body{
```

```
      font-family: Arial, sans-serif;
```

```
      background-color: #f2f2f2;
```

```
      margin: 0;
```

```
      padding: 20px;
```

```
    }
```

```
    h1{
```

```
      text-align: center;
```

```
      color: #333;
```

```
    }
```

```
    .container{
```

```
      display: flex;
```

```
      justify-content: center;
```

```
      gap: 30px;
```

```
      flex-wrap: wrap;
```

```
    }
```

```
.form-box {  
    width: 400px;  
    background-color: white;  
    padding: 25px;  
    border: 2px solid #ccc;  
    border-radius: 10px;  
    box-shadow: 0 0 10px gray;  
}
```

```
h2 {  
    text-align: center;  
    color: #444;  
}
```

```
label {  
    font-weight: bold;  
    display: block;  
    margin-top: 12px;  
    margin-bottom: 5px;  
}
```

```
input[type="text"],  
input[type="email"],  
input[type="password"],  
input[type="date"],
```

```
select,  
textarea {  
    width: 100%;  
    padding: 10px;  
    border: 1px solid #aaa;  
    border-radius: 5px;  
    box-sizing: border-box;  
    font-size: 14px;  
}
```

```
textarea {  
    resize: vertical;  
}
```

```
.radio-group,  
.checkbox-group {  
    margin: 8px 0;  
}
```

```
.radio-group label,  
.checkbox-group label {  
    display: inline;  
    font-weight: normal;  
    margin-right: 10px;  
}
```

```
.buttons {  
    text-align: center;  
    margin-top: 20px;  
}
```

```
input[type="submit"],  
input[type="reset"] {  
    padding: 10px 25px;  
    border: none;  
    border-radius: 5px;  
    margin: 5px;  
    font-size: 15px;  
    cursor: pointer;  
}
```

```
input[type="submit"] {  
    background-color: green;  
    color: white;  
}
```

```
input[type="reset"] {  
    background-color: red;  
    color: white;  
}
```

```
input[type="submit"]:hover,
```

```
        input[type="reset"]:hover {
            opacity: 0.8;
        }
    </style>
</head>

<body>

    <h1>College Portal</h1>

    <div class="container">

        <!-- Login Form -->
        <div class="form-box">

            <h2>Login Form</h2>

            <form>

                <label for="username">Username:</label>
                <input type="text" id="username"
                    placeholder="Enter username">

                <label for="login-email">Email:</label>
                <input type="email" id="login-email"
                    placeholder="Enter email">
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password"  
      placeholder="Enter password">
```

```
<div class="buttons">
```

```
  <input type="submit" value="Login">
```

```
  <input type="reset" value="Reset">
```

```
</div>
```

```
</form>
```

```
</div>
```

```
<!-- College Admission Form -->
```

```
<div class="form-box">
```

```
<h2>College Admission Form</h2>
```

```
<form>
```

```
<label for="name">Student Name:</label>
```

```
<input type="text" id="name"  
      placeholder="Enter your name">
```

<label for="email">Email:</label>

<input type="email" id="email"
placeholder="Enter your email">

<label for="dob">Date of Birth:</label>

<input type="date" id="dob">

<label>Gender:</label>

<div class="radio-group">

<input type="radio" name="gender" value="male">

<label>Male</label>

<input type="radio" name="gender" value="female">

<label>Female</label>

<input type="radio" name="gender" value="other">

<label>Other</label>

</div>

<label for="course">Select Course:</label>

<select id="course">

<option value="">-- Select Course --</option>

<option value="bca">BCA</option>

<option value="bba">BBA</option>

```
<option value="bcom">B.Com</option>
<option value="btech">B.Tech</option>
</select>

<label>Hobbies:</label>

<div class="checkbox-group">
  <input type="checkbox" name="hobby" value="reading">
  <label>Reading</label>

  <input type="checkbox" name="hobby" value="coding">
  <label>Coding</label>

  <input type="checkbox" name="hobby" value="sports">
  <label>Sports</label>
</div>

<label for="address">Address:</label>
<textarea id="address" rows="4"
  placeholder="Enter your address"></textarea>

<div class="buttons">
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</div>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

4. JavaScript Calculator: Create a simple calculator using HTML and JavaScript to perform addition, subtraction, multiplication, and division of two numbers.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Simple Calculator</title>
```

```
<style>
```

```
body {  
    font-family: Arial, sans-serif;  
    background-color: #f2f2f2;  
    text-align: center;  
    padding: 50px;  
}
```

```
.calculator {  
    width: 350px;  
    margin: auto;
```

```
padding: 25px;
background-color: white;
border: 2px solid #333;
border-radius: 10px;
}
```

```
input {
width: 90%;
padding: 10px;
margin: 10px;
font-size: 16px;
}
```

```
button {
padding: 10px 15px;
margin: 5px;
font-size: 16px;
cursor: pointer;
}
```

```
#result {
font-size: 20px;
font-weight: bold;
margin-top: 20px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="calculator">
```

```
<h1>Simple Calculator</h1>
```

```
<input type="number" id="num1" placeholder="Enter first number">
```

```
<input type="number" id="num2" placeholder="Enter second number">
```

```
<br>
```

```
<button onclick="add()">+</button>
```

```
<button onclick="subtract()">-</button>
```

```
<button onclick="multiply()">×</button>
```

```
<button onclick="divide()">÷</button>
```

```
<p id="result">Result: </p>
```

```
</div>
```

```
<script>
```

```
function getNumbers() {
```

```
    let a = parseFloat(document.getElementById("num1").value);  
    let b = parseFloat(document.getElementById("num2").value);  
  
    return [a, b];  
}
```

```
function add() {  
    let [a, b] = getNumbers();  
    document.getElementById("result").innerHTML =  
        "Result: " + (a + b);  
}
```

```
function subtract() {  
    let [a, b] = getNumbers();  
    document.getElementById("result").innerHTML =  
        "Result: " + (a - b);  
}
```

```
function multiply() {  
    let [a, b] = getNumbers();  
    document.getElementById("result").innerHTML =  
        "Result: " + (a * b);  
}
```

```
function divide() {  
    let [a, b] = getNumbers();
```

```
    if (b == 0) {  
        document.getElementById("result").innerHTML =  
            "Cannot divide by zero!";  
    } else {  
        document.getElementById("result").innerHTML =  
            "Result: " + (a / b);  
    }  
}
```

```
</script>
```

```
</body>
```

```
</html>
```

5. JavaScript Validation: Create a Student Registration Form using HTML and JavaScript. Validate fields such as Name, Email, Mobile Number, Password, and Age before submitting the form. Display appropriate validation messages.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Student Registration Form</title>
```

```
    <style>
```

```
        body {  
            font-family: Arial, sans-serif;  
            background-color: #f2f2f2;  
        }
```

```
.container {  
    width: 450px;  
    margin: 40px auto;  
    padding: 25px;  
    background-color: white;  
    border: 2px solid #333;  
    border-radius: 10px;  
}
```

```
h1 {  
    text-align: center;  
}
```

```
label {  
    font-weight: bold;  
}
```

```
input {  
    width: 100%;  
    padding: 10px;  
    margin: 8px 0;  
    box-sizing: border-box;  
}
```

```
.error {
```

```
    color: red;
    font-size: 14px;
}
```

```
input[type="submit"],
input[type="reset"] {
    width: 48%;
    background-color: green;
    color: white;
    border: none;
    cursor: pointer;
}
```

```
input[type="reset"] {
    background-color: red;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>Student Registration</h1>
```

```
<form onsubmit="return validateForm()">
```

<label>Name:</label>

<input type="text" id="name">

<div id="nameError" class="error"></div>

<label>Email:</label>

<input type="text" id="email">

<div id="emailError" class="error"></div>

<label>Mobile Number:</label>

<input type="text" id="mobile">

<div id="mobileError" class="error"></div>

<label>Password:</label>

<input type="password" id="password">

<div id="passwordError" class="error"></div>

<label>Age:</label>

<input type="number" id="age">

<div id="ageError" class="error"></div>

<input type="submit" value="Submit">

<input type="reset" value="Reset">

```
</form>
```

```
</div>
```

```
<script>
```

```
function validateForm() {
```

```
    let name = document.getElementById("name").value.trim();  
    let email = document.getElementById("email").value.trim();  
    let mobile = document.getElementById("mobile").value.trim();  
    let password = document.getElementById("password").value;  
    let age = document.getElementById("age").value;
```

```
    let valid = true;
```

```
    // Clear previous messages
```

```
    document.getElementById("nameError").innerHTML = "";  
    document.getElementById("emailError").innerHTML = "";  
    document.getElementById("mobileError").innerHTML = "";  
    document.getElementById("passwordError").innerHTML = "";  
    document.getElementById("ageError").innerHTML = "";
```

```
    // Name validation
```

```
    if (name == "") {  
        document.getElementById("nameError").innerHTML =
```

```
        "Name is required.";
        valid = false;
    } else if (!/^[A-Za-z ]+$/ .test(name)) {
        document.getElementById("nameError").innerHTML =
            "Name should contain only letters.";
        valid = false;
    }
}
```

// Email validation

```
let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (email == "") {
    document.getElementById("emailError").innerHTML =
        "Email is required.";
    valid = false;
} else if (!emailPattern.test(email)) {
    document.getElementById("emailError").innerHTML =
        "Enter a valid email address.";
    valid = false;
}
```

// Mobile number validation

```
if (mobile == "") {
    document.getElementById("mobileError").innerHTML =
        "Mobile number is required.";
    valid = false;
}
```

```
} else if (!/^[0-9]{10}$/.test(mobile)) {  
    document.getElementById("mobileError").innerHTML =  
        "Mobile number must contain exactly 10 digits.";  
    valid = false;  
}  
  
// Password validation  
if (password == "") {  
    document.getElementById("passwordError").innerHTML =  
        "Password is required.";  
    valid = false;  
} else if (password.length < 6) {  
    document.getElementById("passwordError").innerHTML =  
        "Password must be at least 6 characters.";  
    valid = false;  
}  
  
// Age validation  
if (age == "") {  
    document.getElementById("ageError").innerHTML =  
        "Age is required.";  
    valid = false;  
} else if (age < 18 || age > 60) {  
    document.getElementById("ageError").innerHTML =  
        "Age must be between 18 and 60.";  
    valid = false;
```

```
}
```

```
if (valid) {
```

```
    alert("Registration successful!");
```

```
}
```

```
return valid;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```